

Youssif Sameh Abdelrady Mohamed

AI / Machine Learning Engineer · Computer & Systems Engineering Student

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PROFESSIONAL SUMMARY

AI/ML Engineering student with hands-on experience in machine learning, generative AI, and data analysis. Skilled in building end-to-end AI pipelines using Python, Scikit-learn, and deep learning frameworks. Experienced through internships at DEPI and NTI, with a strong foundation in algorithms, OOP, and software engineering. Seeking AI/ML Engineering opportunities to build scalable and impactful intelligent systems.

EDUCATION

Capital University

Sep 2022 – Jun 2027

Bachelor's Degree in Computer and Systems Engineering

- Relevant Coursework: Data Structures & Algorithms, OOP, Database Systems, Digital Signal Processing, Embedded Systems

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, MATLAB, SQL

AI / ML & Data: Machine Learning, Deep Learning, Generative AI, Data Analysis, NumPy, Pandas, Scikit-learn, TensorFlow

Embedded & IoT: AVR Microcontrollers, Arduino, IoT Protocols (MQTT), Sensor Integration

Tools & Platforms: Git, Linux, Windows, SolidWorks, Jupyter Notebook, VS Code, Google Colab

Concepts: OOP, Data Structures & Algorithms, Computer Vision, Agile/Scrum

EXPERIENCE

Generative AI Intern

Nov 2025 – Present

Digital Egypt Pioneers Initiative (DEPI) · Cairo, Egypt

- Developed and fine-tuned generative AI models for real-world applications using Python, TensorFlow, and PyTorch.
- Built AI-powered pipelines to automate data processing, applying hyperparameter tuning and evaluation techniques.
- Collaborated with cross-functional teams to deliver scalable and efficient AI solutions.

Data Analysis Intern

Oct 2025 – Jan 2026

National Telecommunication Institute (NTI) · Cairo, Egypt

- Analyzed real-world datasets using Python, Pandas, and NumPy to extract insights and support data-driven decisions.
- Created visualizations and reports, and automated data cleaning tasks to improve workflow efficiency.
- Applied EDA and statistical analysis techniques to identify trends and patterns in datasets.

PROJECTS

MalwareFlow — LLM-Powered Malware Analysis Pipeline

2026

Personal Project · github.com/youssifsameh1000/MalwareFlow

- Built a full-stack malware analysis platform integrating VirusTotal, CAPA, and LLM reasoning to generate automated threat reports.
- Developed a concurrent FastAPI backend with modular pipeline architecture; implemented PE parsing, entropy analysis, IOC detection, and MITRE ATT&CK mapping.
- Created a React + Vite dashboard with real-time analysis tracking and structured reporting.

Minimax Pacman Ghost Agent — Adversarial AI

2026

Academic Project · github.com/youssifsameh1000/Minimax-Pacman-Ghost-Project

- Developed AI ghost agents using Minimax and Alpha-Beta Pruning for the UC Berkeley CS188 Pacman framework.
- Implemented recursive and iterative adversarial search algorithms with benchmarking and evaluation functions.

Bicycle Shop Sales Analysis

2025

End-to-End Data Analysis · github.com/youssifsameh1000/bicycle-shop-sales-analysis

- Performed data cleaning, EDA, and SQL analysis on relational sales datasets; built interactive Power BI dashboards with KPIs for sales, profit, and customer trends.

LEADERSHIP & ACTIVITIES

Robotics Team Member · Enjad Student Organization

Jul 2024 – Mar 2025

- Contributed to design and programming of autonomous robotic systems, integrating mechanical and software components.

Technical Support (RAS) · IEEE Helwan Student Branch

Sep 2023 – Aug 2024

- Delivered technical workshops on embedded systems and programming to fellow students.

Media Team Member · FEHU-CPC

Sep 2024 – Aug 2025

- Designed event posters, social media graphics, and promotional videos using Adobe Photoshop to boost engagement.

CERTIFICATIONS & TRAINING

- DEPI Generative AI · NTI Data Analysis · ITI AI Program
- ITI IoT · ODC Cybersecurity · AMIT Embedded Systems

LANGUAGES

Arabic: Native | English: Advanced (Professional Working Proficiency)